MULTI-AGENT ELECTION LAB · PART 2

Who steers the winner on purpose

Four deliberate manipulations, each run against an untouched twin. Who wins, how, and who can catch them.

Team Eclipse · Experiments 5 to 8 · towns of 100 people, same engine

THE QUESTION

If someone tries to steer the result on purpose, does it work, and can anyone catch them?

EXPERIMENT 5

A watchdog that sees only totals

10 attack styles · 420 runs

EXPERIMENT 6

Foreign spies

15 sleeper cells · 4 tactics × 8 towns

EXPERIMENT 7

Changing minds slowly

4 campaigns, same budget × 5 towns

EXPERIMENT 8

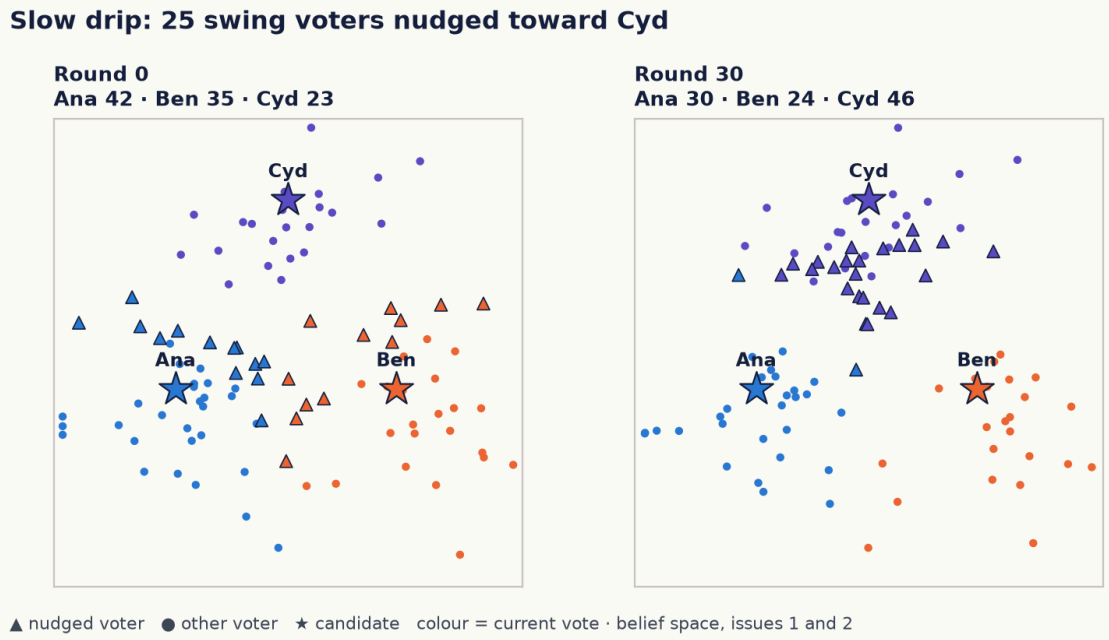
Campaigns vs echo chambers

3 parties × 5 playbooks: 125 match-ups × 3 towns

Every attack run is paired with a twin town that is identical except that the attacker stays quiet

SETUP FOR EXPERIMENT 5

A watchdog that sees only the scoreboard



Attack W4, slow drip. Every round the attacker nudges the beliefs of the 25 voters closest to switching a little toward Cyd. By round 30 Cyd has gone from third (23) to first (46).

THE TOWN

100 people with beliefs on two issues. Three candidates: **Ana 42, Ben 35, Cyd 23**. Everyone votes for the nearest candidate; beliefs also drift a little by themselves.

THE ATTACKER

A fixed budget to push chosen voters' beliefs toward Cyd. Ten styles: fast or slow, few or many voters, aimed or random.

THE WATCHDOG

Sees only each round's vote totals. It **wins** if it raises the alarm before Cyd wins three rounds in a row.

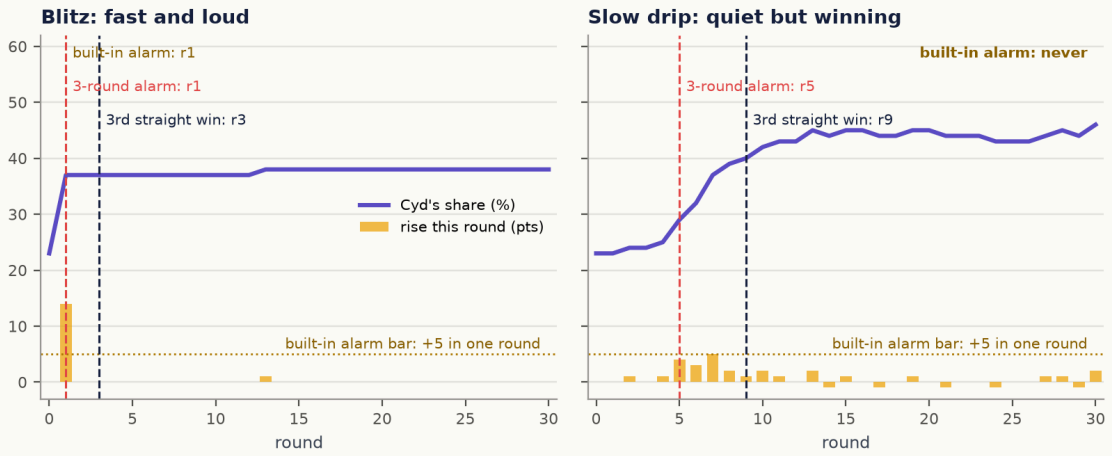
BUILT-IN ALARM

Fires when the target jumps more than 5 points in one round.

Belief drift up to 0.01 per round · 30 rounds · every attack paired with an untouched twin town

EXPERIMENT 5

Slow attacks slip under a one-round alarm



Purple: Cyd's share. **Yellow bars:** its rise each round. A blitz jumps 14 points at once; the slow drip never rises more than 5 in a round, yet wins by round 9.

WHAT WE CHANGED

Ten ways to spend the attack budget, one town each, each with an untouched twin.

WHAT WE GOT

The built-in alarm catches all 3 fast attacks, but **misses both slow drips**, which still win by round 9 and 10. Five attacks never win: wrong targets, random pushes, too little money, too few or too many voters.

WHY

A drip adds 4 to 5 points a round: always under the one-round bar, but it keeps adding up.

OUR REAL-WORLD READING

Watching for sudden jumps misses patient campaigns.

No alarm fired in any of the 10 twin towns · the attack only wins on the 25 voters closest to switching

EXPERIMENT 5 · BEST STRATEGIES

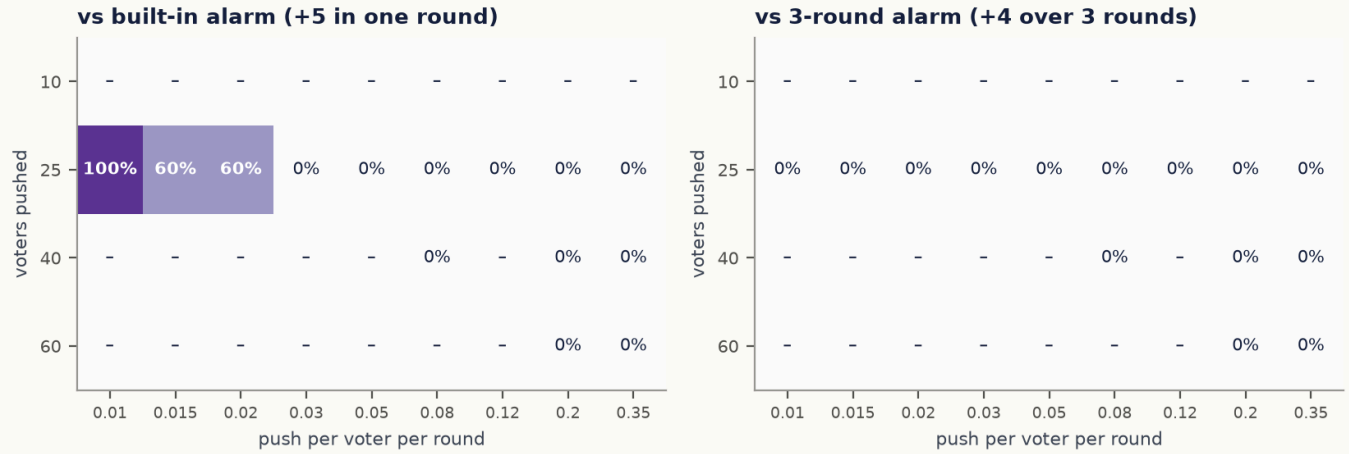
Look back three rounds, not one

20 of 25
attacks caught by the built-in alarm

25 of 25
caught by the 3-round alarm

0 of 40
false alarms in untouched towns

0 of 36
attack plans that win unseen



Cell: share of 5 test towns where the attacker wins without being caught first · – = attack never wins

THE WATCHDOG'S BEST RULE

We tried 101 rules that use only vote totals and kept the fastest one that never fired in 80 untouched towns, even with 50% more natural drift: **alarm if the target rose more than 4 points over the last 3 rounds**.

THE ATTACKER'S BEST REPLY

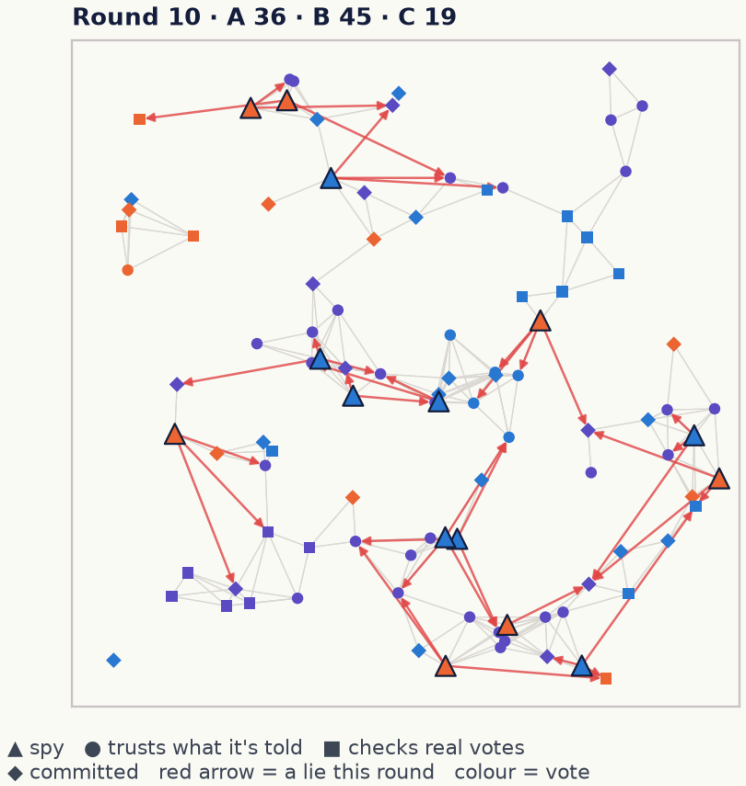
Against the one-round alarm: a slow drip on the 25 swing voters wins **every time, unseen**, around round 17. Against the 3-round rule nothing slips through; the best left is a blitz that is flagged in round 1.

Rule chosen on towns 201–205, tested on new towns 101–105 · lowering the one-round bar to 3 points also catches 25 of 25

SETUP FOR EXPERIMENT 6

Sleeper cells in the group chats

Spy cells with a dossier, one town



THE TOWN

85 residents: **A 40%, B 37%, C 23%**, chatting with neighbours and in 10 group chats. Three kinds of resident:

- ◆ **Committed** 40%: always vote their favourite.
- ● **Trusting** 40%: vote what most contacts say they vote.
- ■ **Checkers** 20%: vote what most neighbours actually vote.

THE SPIES · 15%

15 sleepers (▲) vote like locals to blend in and send fake “I'm voting B” messages: 4 a round, 100 each, to their 15 followers and one group chat.

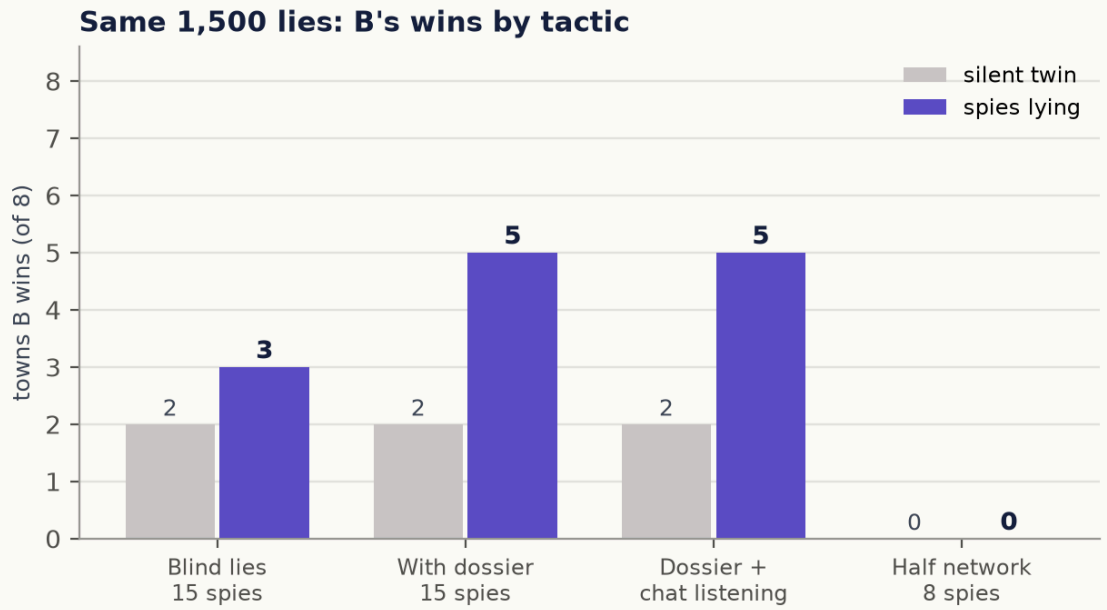
THEIR INTELLIGENCE

None (lie at random), a **dossier** of their followers' true favourites, or the dossier plus **listening** to two nearby group chats.

Each spy is its own cell with its own budget · 25 rounds · every run paired with a twin whose spies stay silent

EXPERIMENT 6

Knowing whom to lie to is the weapon



8 towns per tactic. Grey: the twin town where the spies stay silent. Purple: the same town with the spies lying. Every tactic sends the same 1,500 lies.

WHAT WE CHANGED

Only what the spies know: nothing, a dossier, dossier plus chat listening; then half the network.

WHAT WE GOT

B wins 2 of 8 towns with silent spies, **3 with blind lies, 5 with a dossier**. Listening to chats adds nothing. With 8 spies B never leads.

WHY

Trusting voters break ties with their own favourite, so hidden B fans who now follow A tip with one or two lies. Blind lies land on committed voters and checkers, who never move.

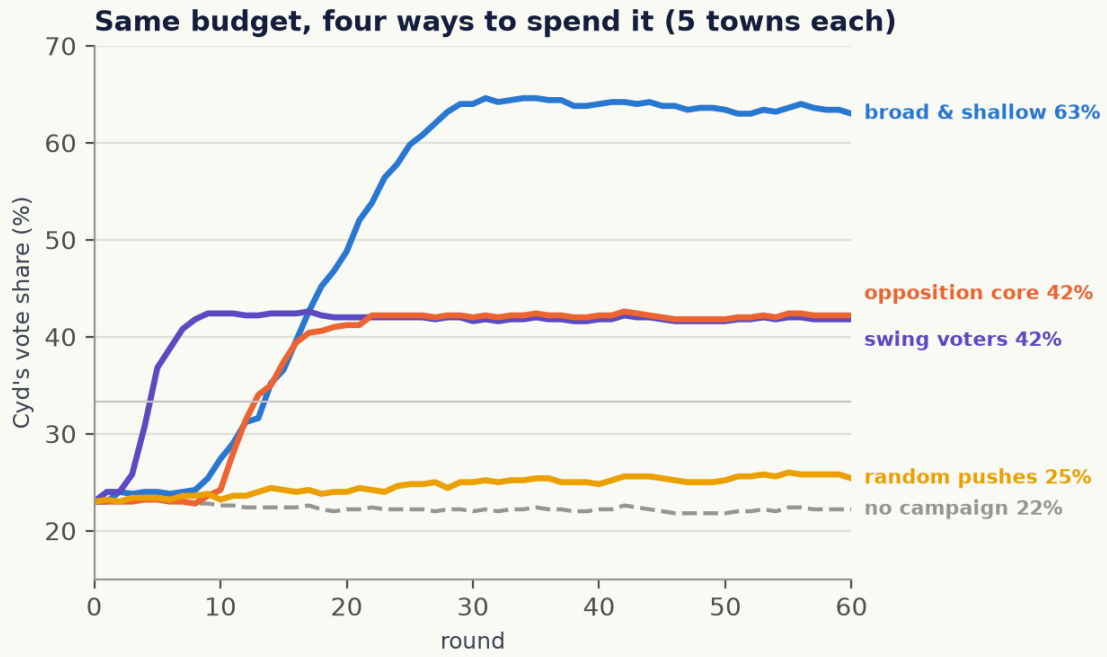
OUR REAL-WORLD READING

Targeting data, not volume, is what makes disinformation work; voters who check what people do, not what they say, are nearly immune.

B's share gain over the silent twin: blind +2.0, dossier +4.8 points · the 8-spy town also starts weaker for B (23% vs 36% in its twin)

EXPERIMENT 7

Changing minds slowly



Same town as Experiment 5, 60 rounds. Each campaign gets the same budget for pushing beliefs toward Cyd and spends it in a different way. Mean of 5 towns.

WHAT WE CHANGED

Where the budget goes: a little on 60 voters, a lot on the 20 swing voters, on 20 of the opposition's core, or in random directions.

WHAT WE GOT

Swing voters: Cyd leads by round 5 and wins 56 of 61 rounds.
Broad and shallow: slower, but 63% of the vote. The core works slowly; random pushes do nothing. The gains stay after the money stops.

WHY

Beliefs drift at random with no pull back, so moved voters stay moved.

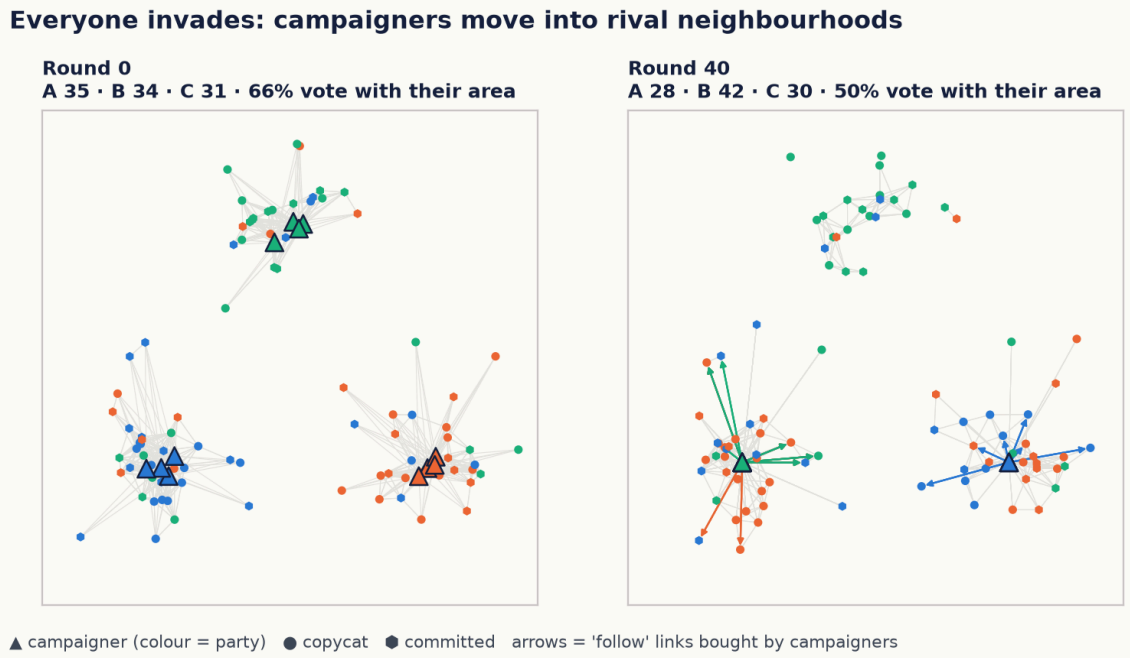
OUR REAL-WORLD READING

The winner ends up 24% further from what people first believed, yet looks closer to what they believe now: successful persuasion hides itself.

Average distance from the winner to voters' original beliefs: 0.436 with any successful campaign vs 0.352 without one

SETUP FOR EXPERIMENT 8

Campaigners who reshape who you hear



All three parties invade. Campaigners walk into rival neighbourhoods and make voters follow them. By round 40 only 50% still vote with their neighbourhood, and B has taken the lead.

THE TOWN

88 residents in three neighbourhoods: **A 34, B 30, C 24**, 65% loyal to the local party. Copycats (60%) hear only 4 or 5 neighbours; the rest are committed.

THE CAMPAIGNERS

4 per party. They walk 6× faster, poll everyone within 0.3, and can make up to 6 voters follow them, 2 new a round.

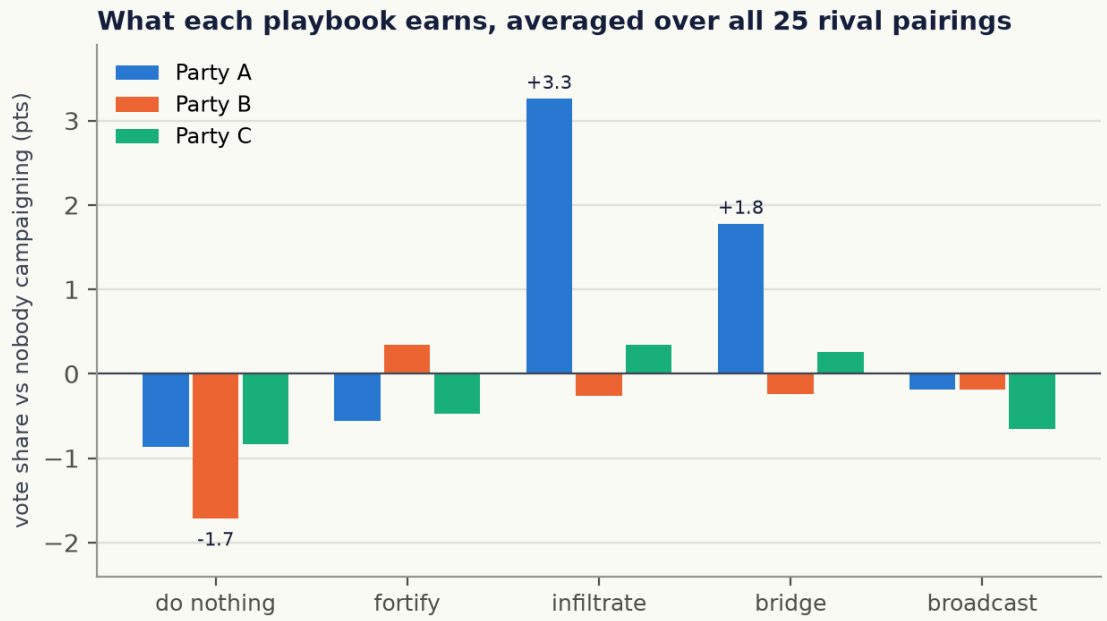
FIVE PLAYBOOKS

Do nothing · **fortify**: stay home, win back defectors · **infiltrate**: move into the biggest rival area · **bridge**: sit on the border, chase swing voters · **broadcast**: random followers.

Every combination of playbooks, 125 in all, × 3 towns × 40 rounds · a party's score: its share over the last 10 rounds

EXPERIMENT 8

The leader attacks, the challenger defends



Each bar: a party's average gain from one playbook, over all 25 combinations of its rivals' playbooks.

WHAT WE CHANGED

Every party's playbook, in every combination.

WHAT WE GOT

A should infiltrate: best in 19 of 25 match-ups, +3.3 points. **B** should fortify; doing nothing costs it 1.7. **C** barely moves: +0.3 at best, and wins **0 of 375** elections.

WHY

Invading the biggest rival splits its copycats. The smallest party has no area big enough to tip.

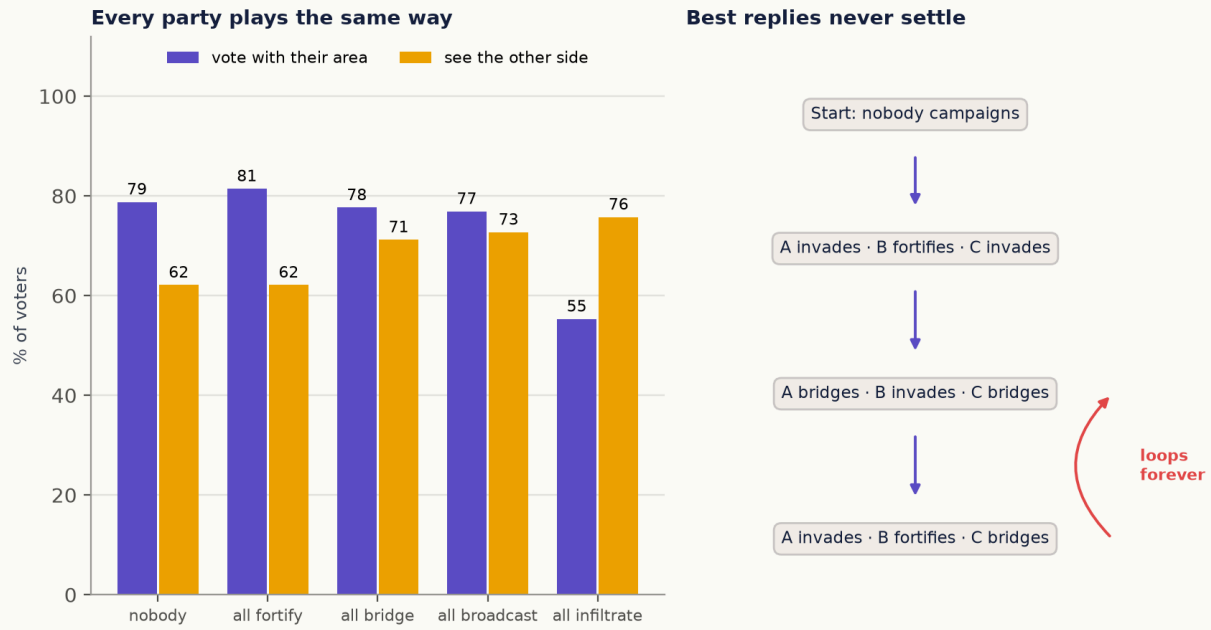
OUR REAL-WORLD READING

B's one route to an upset is to work the border while A stays home: then it wins 2 of 3 towns.

Across all 375 runs: A wins 348, B 27, C 0 · averages over 3 towns per match-up

EXPERIMENT 8 · BEST STRATEGIES

No resting point, only an arms race



Left: every party plays the same playbook. **Right:** each party in turn switches to its best reply; the switching never stops.

WHAT WE EXPECTED

One set of playbooks where no party wants to switch.

WHAT WE GOT

There is none. The best replies loop forever: A swaps invading for bridging, B swaps fortifying for invading. Closest to rest: **everyone invades**; no party gains even a quarter of a vote by switching.

OUR REAL-WORLD READING

When every side campaigns in its rivals' areas, echo chambers break (79% → 55% vote with their neighbourhood; 62% → 76% see the other side), but the winner barely changes.

Most stable mixes: A bridges, B and C invade (best switch gains 0.20 points) and everyone invades (0.23)

TAKEAWAYS · PART 2

5. A one-round alarm misses patient attackers: a 3-round look-back caught 25 of 25, with no false alarms.

6. Targeting beats volume: the same lies take B from 2 to 5 of 8 towns once spies know whom to lie to.

7. Slow persuasion sticks and hides itself: the winner ends up 24% further from what people first believed.

8. Campaign wars break echo chambers but not the winner: no stable playbook, and the underdog won 0 of 375.